

Bayesian Machine Learning Course Program

Weekly overview of topics, videos, and reading material

Week 1

Program

- Introduction to Bayesian inference and the course
- Basic concepts: prior, likelihood, and posterior distributions
- The Beta-binomial model for estimating proportions
- Conjugate models and functional forms
- Simple A/B testing using the Beta-binomial model

Videos

- Lecture1_part1: Introduction to the course
- Lecture1_part2: Bayesian Machine Learning in a nutshell
- Lecture1_part3: Recap of terminology and notation from probability theory
- Lecture1_part4: The binomial distribution and maximum likelihood
- Lecture1_part5: Bayesian inference
- Lecture1_part6: Introduction to exercise sessions

Parts 1 and 2 introduce the course and Bayesian machine learning. Parts 3 and 4 are a warm-up for part 5 and may be faster paced. Part 5 introduces the first Bayesian analysis of the Beta-binomial model.

Reading Material

- The Beta distribution: Murphy1, Section 2.7.4, pages 63–64
- Likelihood functions and MLE for Bernoulli: Murphy1, Sections 4.0–4.2.3, pages 107–111
- Section 4.2.2 is technical and can be skipped
- Poor man's Bayes and MAP estimation for Bernoulli: Murphy1, Sections 4.5–4.5.1, pages 120–122
- Prior, likelihood, posterior, and evidence: Murphy1, Sections 4.6.0–4.6.1, pages 129–130
- The Beta-binomial model: Murphy1, Sections 4.6.2, 4.6.6, 4.6.7, pages 129–137
- Credible intervals: Murphy1, Section 4.6.6, pages 146–147

Week 2

Program

- Foundations of probabilistic machine learning
- Plugin approximations and Dirac's delta distribution
- Grid approximations
- Marginalization and posterior predictive distributions
- Applications to dose-response modelling

Videos

- Lecture2_part1: Recap of central points from last week, 6 min
- Lecture2_part2: Probabilistic machine learning, 21 min
- Lecture2_part3: The plugin approximation, 12 min
- Lecture2_part4: Grid approximations, 19 min
- Lecture2_part5: Introduction to exercise, 17 min

Reading

- Sets of related random variables, sum/product rule, marginalization: Murphy1, Section 2.2.3, pages 38–39
- Independence and conditional independence: Murphy1, Section 2.2.4, pages 39–40
- Bernoulli, Binomial distributions, and logistic regression: Murphy1, Section 2.4, pages 49–53
- Univariate Gaussian distribution: Murphy1, Sections 2.6–2.6.2, pages 57–59
- Dirac's delta: Murphy1, Section 2.6.5, pages 60–61
- Bayesian machine learning: Murphy1, Sections 4.6.7–4.6.8.1, pages 147–152

Week 3

Program

- Linear Gaussian systems and Bayesian linear regression
- Hyperparameter estimation via the marginal likelihood
- The evidence approximation

Videos

- Lecture3_part1: Recap of central concepts from last week, 6 min
- Lecture3_part2: Recap of multivariate Gaussians, 14 min
- Lecture3_part3: Recap of linear regression and supervised learning, 17 min
- Lecture3_part4: Bayesian linear regression, 15 min
- Lecture3_part5: The posterior predictive distribution, 15 min
- Lecture3_part6: Dealing with hyperparameters, 13 min

Parts 1–3 may be skipped if you are confident with the concepts. Parts 4–6 contain the core new material. Linear Gaussian systems are strongly recommended.

Reading

- Joint distributions: Murphy1, Section 3.1, pages 77–80
- Multivariate Gaussian distribution: Murphy1, Sections 3.2–3.2.3, pages 80–84
- Linear Gaussian systems: Murphy1, Section 3.3, pages 86–92
- Linear regression recap: Murphy1, Sections 11–11.2.2.1, pages 369–371
- MLE for noise variance: Murphy1, Section 11.2.3.6, page 377
- Ridge regression and MAP estimation: Murphy1, Sections 11.3–11.3.1, pages 379–380
- Bayesian linear regression: Murphy1, Section 11.7, pages 403–409
- Model comparison and marginal likelihood: Bishop, Sections 3.4–3.5, pages 161–172

Week 4

Program

- Classical/frequentist statistics vs Bayesian statistics
- Introduction to Bayesian methods for classification
- Bayesian logistic regression for classification
- The Laplace approximation

Videos

- Lecture 4 part 1: Summary of last week and graphical models, 20 min
- Lecture 4 part 2: Bayesian statistics vs classical statistics, 5 min
- Lecture 4 part 3: Bayesian methods for classification, 20 min
- Lecture 4 part 4: Bayesian logistic regression for classification, 8 min
- Lecture 4 part 5: The Laplace approximation, 10 min
- Lecture 4 part 6: Evaluating the posterior predictive distribution, 13 min

Readings

- Graphical models: Section 3.6, pages 99–103
- Credible intervals: Section 4.6.6, pages 146–147
- Confidence intervals: Section 4.7.4, pages 157–158
- Credibility vs confidence intervals: Section 4.7.5, pages 158–159
- Binary logistic regression: Section 10.1–10.2.4, pages 337–343
- Bayesian logistic regression and Laplace approximation: Section 10.5, pages 361–366

Week 5

Program

- Distributions on function spaces
- Kernels and covariance functions
- Gaussian processes for regression

Videos

- Lecture5 part 1: From parameter space to function space
- Lecture5 part 2: A visual introduction
- Lecture5 part 3: Gaussian process regression
- Lecture5 part 4: Covariance functions
- Lecture5 part 5: Hyperparameters

Reading

- Marginals and conditionals of a multivariate Gaussian: Section 3.2.3, page 84
- Kernels: Section 17–17.1, pages 565–572
- Gaussian processes: Section 17.2–17.2.6.2, pages 572–578
- *Section 17.2.3 can be skipped*
- Empirical priors: Section 4.6.5.3, pages 145–146
- Stationary kernels: Murphy2, Sections 18.2.1–18.2.1.3, pages 676–678

Week 6

Program

- More on kernel functions
- Gaussian processes for classification
- Neural networks for probabilistic modelling

Videos

- Lecture 6 part 1: More on covariance functions, 17 min
- Lecture 6 part 2: Gaussian process software and tools, 3 min
- Lecture 6 part 3: Gaussian processes for binary classification, 22 min
- Lecture 6 part 4: Neural networks for probabilistic modelling, 11 min

Reading

- Murphy2: Sections 18.4–18.4.2, pages 701–703
- Murphy1: Sections 13.1–13.2.4, pages 423–432

There is a typo in the equation for expected calibration error in the book and video, but it is corrected in the exercise and slides.

Week 7

Program

- Connections between Gaussian processes and neural networks
- Models with non-Gaussian likelihoods
- Generalization error
- Decision theory
- Entropy and calibration

Videos

- Lecture 7 part 1: GP and NNs
- Lecture 7 part 2: Non-Gaussian likelihoods

- Lecture 7 part 3: Defining the generalization error
- Lecture 7 part 4: Statistical/Bayesian decision theory
- Lecture 7 part 5: Calibration for classification models

Reading

- Common univariate distributions: Murphy1, Section 2.7, pages 61–65
- Generalized linear models: Murphy1, Section 12.1, pages 413–415
- Gaussian processes with non-Gaussian likelihoods: Murphy2, Section 18.4, pages 693–697
- Bayesian decision theory: Murphy1, Sections 5.1–5.1.5, pages 167–177
- Evaluating predictive models and calibration: Murphy2, Section 14.2, pages 572–576
- Entropy: Murphy1, Section 6.1, pages 205–206

Week 8

Program

- More on calibration
- Monte Carlo methods
- Ancestral sampling, rejection sampling, importance sampling, and transformations
- MCMC and the Metropolis algorithm

Videos

- Lecture 8 part 1: More on calibration
- Lecture 8 part 2: Monte Carlo methods

- Lecture 8 part 3: Simple MC methods
- Lecture 8 part 4: MCMC and Metropolis algorithm

Reading

- Monte Carlo methods: Murphy2, Sections 11–11.5.3, pages 477–486
- Markov chain Monte Carlo: Murphy2, Sections 12–12.2.3.3, pages 492–498
- Optional additional resource: Bishop, pages 523–542

Week 9

Program

- More MCMC theory
- Hamiltonian Monte Carlo
- ULA and MALA algorithms
- MCMC convergence diagnostics

Videos

- Part 1: More theory for MCMC, 16 min
- Part 2: Introduction to Hamiltonian Monte Carlo, 10 min
- Part 3: Hamiltonian Dynamics, 23 min
- Part 4: Hamiltonian Monte Carlo, 15 min
- Part 5: ULA and MALA algorithms, 8 min
- Part 6: Convergence diagnostics, 10 min

Reading

- Stationary distribution of a Markov chain: Murphy2, Section 2.6.4, pages 51–55
- Metropolis-Hastings: Murphy2, Section 12.2, pages 494–497
- Hamiltonian Monte Carlo: Murphy2, Section 12.5, pages 510–518
- MCMC convergence: Murphy2, Section 12.6, pages 518–526

Week 10

Program

- Wrap up MCMC and sampling methods
- Introduction to variational inference
- Variational families and KL divergence
- Coordinate-ascent variational inference
- Mixture models and the Dirichlet distribution
- Bayesian inference for Gaussian mixture models

Videos

- Lecture 10 part 1: Wrapping up MCMC
- Lecture 10 part 2: Intro to VI and KL divergence
- Lecture 10 part 3: Minimizing KL divergence via the ELBO
- Lecture 10 part 4: The CAVI algorithm
- Lecture 10 part 5: Mixture models and the Bayesian GMM

Reading

- Student's t distribution: Murphy2, Section 2.2.2.3, pages 11–12
- Wishart distribution: Murphy2, Section 2.2.5.5, page 19

- Dirichlet distribution: Murphy2, Section 2.2.5.7, pages 20–22
- Mixture models: Murphy2, Sections 28–28.2.1, pages 919–922
- KL divergence: Murphy2, Section 2.7, pages 55–56
- Intro to variational inference: Murphy2, Sections 10–10.1.3, pages 433–438
- Coordinate Ascent VI: Murphy2, Sections 10.3–10.3.5, pages 449–457
- VI for GMM: Murphy2, Sections 10.3.5–10.3.6.6, pages 457–464

Some material from earlier course editions has been restructured. Sections on inducing point methods and sparse Gaussian processes are not part of the curriculum and can be skipped.

Week 11

Program

- More on variational inference and KL divergence
- Free-form vs fixed-form variational inference
- *Hyperparameter estimation in VI*
- *Alpha-divergences*
- Black-box variational inference
- Gradient estimators and mini-batching

Videos

- Lecture 11 part 1: Interpretation of the ELBO, 7 min
- Lecture 11 part 2: Free-form vs fixed-form, 12 min
- *Lecture 11 part 3: Hyperparameter estimation, 5 min*
- Lecture 12 part 1: KL divergences and alpha-divergences, 15 min

- *Lecture 12 part 2: KL divergence and MLE, Information gain, 8 min*
- Lecture 12 part 3: Introduction to BBVI, 9 min
- Lecture 12 part 4: Score function gradient estimator, 12 min
- Lecture 12 part 5: BBVI algorithm, 12 min
- Lecture 12 part 6: Stochastic optimization, 12 min
- Lecture 12 part 7: Reparameterized gradient, 8 min
- Lecture 12 part 8: Mini-batching, 5 min

Reading

- Free-form vs fixed-form VI: Murphy2, Section 10.1.2, pages 441–442
- *Hyperparameter estimation using VI: Murphy2, Section 10.1.3, page 442*
- *Relation between KL and MLE: Murphy2, Section 5.1.6, pages 230–231*
- Entropy: Murphy2, Section 5.2, pages 234–236
- Score function gradient estimator: Murphy2, Sections 6.3.3–6.3.4, pages 273–275
- Reparameterization trick: Murphy2, Sections 6.3.5–6.3.5.2, pages 275–277
- Black-box VI: Murphy2, Sections 10.2–10.2.3.2, pages 445–455

Week 12

Program

- Wrapping up VI
- *Quick discussion of variational autoencoders*
- Bayesian neural networks
- Motivation and challenges
- Monte Carlo Dropout

- Last-layer Laplace approximations
- Deep ensembles

Videos

- *Lecture 12 part 1: Variational autoencoders, 13 min*
- Lecture 12 part 2: Summary of VI, 7 min
- Lecture 12 part 3: Introduction to BNNs, motivation, challenges, and priors, 24 min
- Lecture 12 part 4: Approximating the posterior of BNNs and Dropout, 12 min
- Lecture 12 part 5: Mean-field for BNNs, 7 min
- Lecture 12 part 6: Laplace approximations for BNNs, 17 min
- Lecture 12 part 7: Deep ensembles, 10 min

Reading

- Intro to BNNs: Murphy2, Chapter 17, pages 647–650
- Posteriors of BNNs: Murphy2, Sections 17.3–17.3.9.4, pages 651–660